

Retail Customer Intelligence

Business Analytics & Store Insights — Project Write-up

1. What this project does

This project watches a video of someone shopping in a supermarket and works out a few things a real store manager might want to know: how busy the store was, which parts of the aisle got the most attention, and how long people stuck around. It uses a pretrained YOLO model (yolo26n.pt) to find people in each frame, and ByteTrack to keep track of who is who across frames.

The video I used was filmed from a shopping cart, walking through a few aisles — dry goods, the freezer section, and a fresh produce display. It's about 30 seconds long, 918 frames at 30fps.

2. Detection and tracking (Stage 1)

Every frame, the model looks for people only (class 0 in YOLO) and draws a box around each one. On its own, this doesn't know that the person in frame 1 is the same person in frame 50 — that's what tracking is for. I used `model.track()` with ByteTrack, which gives every detected person an ID number that (mostly) stays the same as they move through the video.

I confirmed this worked by running the full video and checking that boxes with ID numbers showed up correctly on a live preview window before moving on.

3. Zone definitions — the big design decision (Stage 2)

The brief expects zones to represent real store locations, like “Zone A = Dairy Aisle.” That works fine for a fixed CCTV camera that never moves. My video is filmed from a moving cart, so a fixed rectangle of pixels doesn't point at the same physical spot in

the store for very long — it might be the freezer aisle in one frame and the produce section twenty seconds later.

Because of this, I decided to define zones relative to the camera's view instead of the real store layout:

- LEFT — left side of the frame
- CENTER — the middle, roughly where the walking path is
- RIGHT — right side of the frame
- The bottom ~45% of every frame is excluded completely, since that's just my own shopping cart, not real store space.

I checked this visually by drawing the zone lines on actual frames and adjusting the numbers until the lines lined up with the shelf edges and the walking path, rather than just guessing percentages.

```
cart_cutoff_y = frame_height * 0.55
```

```
zone_left_x   = frame_width * 0.30
```

```
zone_right_x  = frame_width * 0.70
```

Limitation I'm aware of: because the camera moves, these zones mean slightly different things depending on how close the camera is to a shelf at any given moment. A wide shot cleanly shows shelf / floor / shelf, but a close-up shot mostly just shows three parts of the same shelf face. I think this is an honest trade-off of using real handheld footage rather than stock CCTV video, not a mistake in the zone logic itself.

4. Occupancy and traffic analytics (Stage 3)

Once zones existed, I needed real numbers, not just a label on a box. Two things mattered here:

- Occupancy — how many people are visible right now, and the highest that number ever got (peak occupancy).
- Unique visitors per zone — counting each track ID only once per zone, using a Python set, instead of counting every single frame someone appears in (which would massively inflate the numbers).

Results from my video: peak occupancy of 4 people at once. Unique visitors: 18 in LEFT, 28 in CENTER, 18 in RIGHT.

5. Dwell time, and a real problem I ran into (Stage 4)

Dwell time is how long each tracked person stayed visible, calculated as (last frame seen – first frame seen) ÷ fps. My first run gave 44 different track IDs for a video where at most 4 people were ever visible at once, and an average dwell time of only 0.6 seconds. That's clearly too fragmented to be real — the tracker was constantly losing people and handing out brand new IDs.

I think this happened mainly because ByteTrack expects a roughly steady camera (like fixed CCTV), and my handheld/cart camera is constantly moving, which confuses its sense of what's consistently moving frame to frame. People also walk behind shelves or get partially blocked, which causes confidence to dip and tracks to break.

To try to fix this, I lowered the detection confidence from 0.4 to 0.3, and wrote a custom tracker config that gives ByteTrack more patience before giving up on a lost person (track_buffer raised from about 30 frames to 60, roughly doubling how long it waits before assuming someone is gone).

```
track_buffer: 60          # was ~30 frames

track_high_thresh: 0.25

match_thresh: 0.8
```

Result: average dwell time improved from 0.6s to 0.9s, and the single longest-tracked person went from 3.2s to 7.5s. But the total number of unique IDs didn't go down (44 → 51) — the lower confidence threshold let in more short, marginal detections that became their own brief new tracks. So this was a partial improvement, not a full fix. I think properly solving this would need a smarter tracker that can recognise a person by appearance, not just by motion, which is outside the scope of what this project asks for.

6. Heatmap (Stage 5)

To show where people spent the most time on screen, I built up a heatmap: every frame, a small circle of “heat” gets added at each detected person's location, and circles that overlap across many frames build up into hot spots. At the end, I turned this into a colour image (blue = low traffic, red = high traffic) and also blended a lighter version live onto the video.

My first attempt at this didn't work properly — it just gave flat solid red blobs on a flat blue background, no real gradient. The cause was twofold: I was painting a flat circle each time instead of properly adding heat values together, and even after fixing that, the colour scaling stretched purely between the lowest and highest values, which with so few people in frame meant almost everything was either completely untouched or very heavily touched, with nothing in between. Applying a square root to the heat values before scaling fixed this, by compressing the high end and spreading out the low-to-medium range, which is where the real variation actually was.

The final heatmap showed one clear hot spot roughly in the centre-left of frame, which matched what the zone numbers were already saying — CENTER was the most visited zone overall.

7. When is the store busiest, and are any zones ignored?

The brief asks two specific questions under the business dashboard section, so I'm answering them directly here with actual data rather than guessing.

When is the store busiest?

I recorded occupancy (how many people were visible) at every single frame and plotted it as a graph over the full 30 seconds. The first 6 seconds show nobody visible at all. From about 6 to 22 seconds, occupancy is spiky but low — bouncing between 0, 1, and occasionally 2-3 people, never sustained. The clearly busiest period is the last third of the clip, from around 22 seconds onward, where occupancy climbs and stays higher, hitting the peak of 4 people twice — once around the 25 second mark and again around 27-28 seconds.

One honest note: the graph is quite jagged rather than smooth, which is partly real (people genuinely walking in and out of frame) and partly tracking noise (a person flickering in and out of detection for a frame or two). A smoothed/rolling-average version would likely show the same overall pattern more clearly, without the noise.



Are any zones completely ignored?

No zone was completely ignored — all three (LEFT, CENTER, RIGHT) had real visitor counts: 18, 28, and 18 respectively. Ranked from most to least popular: CENTER (28) > LEFT (18) = RIGHT (18). CENTER got noticeably more traffic than either side, which makes sense given that CENTER roughly corresponds to the walking path down the aisle, while LEFT and RIGHT correspond to the shelves themselves.

If I were advising a store manager based on this, I'd say: shelf-level engagement (LEFT/RIGHT) is roughly even on both sides, so there's no sign that one side of this particular aisle is being ignored compared to the other. The real insight is that people spend more visual attention on the walking path itself than on either shelf — which could mean product placement directly in the path of movement (e.g. end-of-aisle displays) might get more attention than mid-shelf placement.

Entry and exit counting

The brief asks for an entry/exit count tied to crossing an entrance zone boundary. I used the same line that separates the cart area from the rest of the frame (`cart_cutoff_y`) as this boundary, since that's effectively where a person enters or leaves the camera's visible field of view in this footage.

My first attempt at this was wrong: I originally counted an 'exit' as someone's position moving into the CART zone, but that practically never happens — a real customer's body doesn't walk into the space occupied by my own shopping cart, so exits stayed stuck at zero. I fixed this by redefining an entry as the first frame a `track_id` appears, and an exit as the moment a `track_id` stops appearing in the tracked list for the rest of the video.

Final result: 53 entries and 52 exits. These numbers are close to the total number of unique track IDs (51, see Stage 4), which makes sense given how the counting works. I want to be upfront that this almost certainly overstates the number of real, distinct customers — because of the same ID-switching problem discussed in Stage 4, a lot of these 'entries' are probably the same handful of real people being re-detected as new tracks after being briefly lost by the tracker, rather than genuinely new customers walking into view.

A note on privacy

This kind of system tracks people's movement and time spent in specific areas, even without knowing their names. That's still personal data, and a real deployment of this should be upfront about it — for example, clear signage at the store entrance saying customer movement is being analysed for store-improvement purposes. Footage and tracking data shouldn't be kept longer than needed for the analysis, and should never be combined with other identifying information (like loyalty card or payment data) without separate, explicit consent. For this project, the video used was filmed by me for the purpose of this assignment, but I'd treat this as a real requirement in any actual deployment.

8. Business dashboard and health score (Stage 6)

Everything so far produces numbers, but a store manager won't read a list of track IDs. I added a live dashboard in the corner of the video showing occupancy, peak occupancy, zone breakdown, average dwell time, the most popular zone, and a single combined Business Health Score out of 100.

I decided occupancy should matter most for the health score, since “is my store getting traffic” is usually the first thing a manager would actually ask. The score is built from three parts:

- Occupancy score (50% weight) — peak occupancy compared against a benchmark of 5 people at once being “fully busy.”
- Dwell score (30% weight) — average dwell time compared against a benchmark of 3 seconds being “good engagement.”
- Zone balance score (20% weight) — how evenly spread traffic is across LEFT/CENTER/RIGHT, using the standard deviation of the three counts.

Final result for my video:

Metric	Score
Occupancy score	80 / 100

Dwell score	31 / 100
Zone balance score	78 / 100
Overall Health Score	65 / 100

The low dwell score is worth reading carefully rather than taking at face value: since I already know average dwell time is probably understated because of the tracker ID-switching problem from Stage 4, the real engagement level is likely better than 31/100 suggests. I think this is a fair and honest way to read my own system's output — trusting the numbers, but knowing where they might be wrong.

9. What the system actually delivers

- An annotated output video (output_annotated.mp4) with colour-coded boxes per zone, track IDs, live dwell time, a heatmap overlay, and a live dashboard.
- A standalone heatmap image (zone_heatmap.png).
- An occupancy-over-time graph (occupancy_over_time.png) showing exactly when the store was busiest.
- A live entry/exit counter and a sorted zone popularity ranking, both on the dashboard and in the printed summary.
- Printed end-of-run statistics: peak occupancy, unique visitors per zone, dwell time per person, average dwell time, most popular zone, zone ranking, entry/exit counts, and the Business Health Score breakdown.

10. Honest limitations

- Zones are camera-relative, not fixed store locations, because the footage is from a moving cart rather than fixed CCTV.
- Unique visitor counts and dwell times are likely overestimated in ID count and underestimated in duration, because ByteTrack sometimes loses a person and assigns a new ID rather than recognising them as the same person returning.
- Entry/exit counts (53 entries, 52 exits) inherit this same problem — they almost certainly overstate the true number of distinct customers, for the same reason.
- yolo26n is the lightest YOLO model; small, distant, or partially turned-away people are sometimes missed. A heavier model (yolo26m) would likely catch more of these, at the cost of speed.
- All benchmark numbers in the health score (5 people = full, 3 seconds = good engagement) are reasonable starting assumptions I chose myself, not fixed industry standards — they could be tuned with more real store data.